# Behavior, Not Self-Report: A Unified Evaluation Framework for Interactive Preference Learning via Post-Interaction Decision Consistency

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## Abstract

Interactive preference learning for personalized agents is increasingly studied using LLM-based user simulators. However, current methods are evaluated under heterogeneous goals and incomparable metrics: some optimize simulated user satisfaction, some match textual preference descriptions, and others report task success without isolating whether the agent has actually learned the user's decision preference. In this paper, we argue for a stricter behavioral view: the most faithful operational signal of preference is not what a user says, but how the user decides. We therefore cast interactive preference learning as *fitting a decision preference distribution through interaction*, and we propose a unified evaluation framework based on *post-interaction decision consistency*. Concretely, each user profile induces a decision preference type, represented by an LLM-simulated user agent that repeatedly makes decisions over candidate actions across contexts. An assistant agent interacts with such users, adapts from observed behavior, and is then evaluated on held-out contexts by comparing its decisions against the user's decisions. Training and testing are split by user profile or preference type rather than by context alone, so that generalization explicitly measures adaptation to unseen preferences. We formalize this setting, define decision-level consistency and ranking-based similarity metrics, specify profile-diversity validation procedures to ensure that simulated users induce meaningfully different decision patterns, and propose a benchmark protocol for evaluating whether an interactive learning mechanism can both fit a class of decision preferences and generalize across heterogeneous preference types. Theoretically, we show that exact-match post-interaction decision consistency is Fisher-consistent for deterministic preference fitting, that a proper-scoring extension identifies stochastic decision distributions, that simulator instability induces an additive estimation bias with finite-sample concentration, and that cross-profile generalization error is controlled by coverage of behavior space. The resulting framework is simulator-agnostic, behavior-centered, and directly aligned with the core question of personalized agents: after interaction, does the assistant decide as the user would decide?

## 1 Introduction

Personalized AI assistants are expected to adapt to users over time. In recommendation, planning, tool use, writing support, and task assistance, the central requirement is not merely to complete tasks, but to complete them in ways that are consistent with how a particular user prefers decisions to be made. This requirement has led to a growing literature on interactive preference learning, where

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an assistant agent infers user preferences from dialogue, corrections, choices, and other behavioral traces.

A practical obstacle is that large-scale human experimentation is expensive. As a result, many recent works train and evaluate interactive personalization mechanisms using LLM-based user simulators. Yet the resulting literature suffers from a serious comparability problem. Different papers evaluate different targets: simulated satisfaction, textual preference matching, reward, task completion, tool success, or overall dialogue quality. These signals are not interchangeable, and none by itself provides a unified, objective answer to the central question: *did the assistant actually learn how this user makes decisions?*

This paper advocates a tighter and more behaviorally grounded perspective.

> *The most operationally faithful reflection of preference is not what the user says, but what the user does.*

In other words, preference should be evaluated through *decision behavior*. If a user consistently chooses fast plans over optimal ones, concise answers over detailed ones, or one tool family over another under recurring trade-offs, then those repeated decisions reveal the user's operative preference structure more directly than verbal self-description alone. This viewpoint suggests a unified evaluation target: after interacting with a user, a personalized assistant should make decisions that are consistent with the user's own decisions on new situations.

From this perspective, an LLM simulator is not primarily a proxy for human identity, personality, or naturalness. It is a controllable generator of *decision preference types*. Each profile, role specification, or synthetic user description defines a decision preference type; the corresponding simulated user agent induces a decision rule or decision distribution over contexts. The assistant's learning problem is then to infer that decision pattern from interaction. The evaluation problem is to test whether the learned interaction mechanism can fit such patterns for seen and unseen preference types.

This reframing has three consequences.

First, the object of study becomes the *interactive learning mechanism*, not the simulator's linguistic realism. We do not require the simulator to be a perfect human proxy. We require it to induce diverse, behaviorally distinguishable decision patterns that are stable enough to support evaluation.

Second, train/test splitting should be performed over *user profiles or preference types*, not only over contexts. If different users correspond to different decision preference types, then generalization should mean adaptation to unseen users with unseen behavioral preferences.

Third, evaluation should focus on *decision consistency or similarity* between the assistant agent and the user agent on held-out contexts after interaction. This produces a common metric space in which different interactive preference-learning methods can be compared fairly.

Based on these observations, we propose a unified framework for evaluating interactive preference learning via *post-interaction decision consistency*. We formalize each user as inducing a conditional decision distribution, define the assistant's objective as fitting that distribution through interaction, specify how simulated preference types are constructed and validated, and propose an experimental protocol in which training and testing are split by profile-level preference types. The core scientific question becomes precise:

> *Can an interactive learning mechanism fit a class of decision preference distributions from behavior, and can it generalize to unseen preference types?*

**Contributions.** This paper makes five contributions:

1. We formulate interactive personalization as behavioral decision-distribution fitting rather than textual preference matching.
2. We propose a unified evaluation framework centered on **post-interaction decision consistency**, which compares assistant decisions with user decisions on held-out contexts.
3. We define a **profile-level train/test protocol** in which user profiles or users instantiate decision preference types, making generalization across preferences explicit.

4. We provide theoretical guarantees showing that exact-match decision consistency is Fisher-consistent for deterministic preference fitting, that a proper-scoring extension identifies stochastic decision distributions, that simulator instability contributes an additive estimation bias with finite-sample concentration, and that profile-space coverage controls worst-case cross-profile generalization.
5. We provide a rigorous benchmark design including **preference-diversity validation**, decision-level metrics, ranking-based metrics, and ablations that isolate whether an interaction mechanism truly adapts to heterogeneous decision preferences.

## 2 Related Work

**LLM user simulation.** LLM-based user simulation is increasingly used to train and evaluate interactive agents. Recent studies have also highlighted substantial simulator sensitivity and Sim2Real gaps, showing that simulated users can be unreliable stand-ins for real human behavior (**??**). Our work does not assume that simulated users are faithful replicas of humans. Instead, we use simulation as a controlled means of instantiating diverse decision preference types and evaluating whether an assistant can fit them from interaction.

**Personalized agent evaluation.** Recent personalized-agent benchmarks increasingly study profile-conditioned assistant behavior in task-oriented settings. PersonaLens evaluates personalization in conversational AI assistants with rich user profiles and interaction histories (**?**). ETAPP emphasizes personalized tool-augmented agents and evaluates performance through task completion and tool invocation quality (**?**). ToolSpectrum further studies personalized tool utilization under both user-profile variation and environmental variation (**?**). Our contribution is complementary to these benchmarks: rather than introducing another task-specific success criterion, we identify a unified evaluation target across personalized decision-making settings, namely whether the agent's post-interaction decisions align with the user's own decisions on held-out contexts.

**Interactive and meta-learned adaptation.** Meta-learning studies how a model can rapidly adapt to new tasks from limited data, with MAML being a canonical example (**?**). Our setting is aligned with this view, except that the tasks are not image classes or regression functions but user-specific decision preference types. The benchmark question is therefore not whether a model performs well on a fixed task, but whether its *interactive adaptation mechanism* can infer a new user's decision policy from sparse behavioral evidence.

## 3 Problem Setup

### 3.1 Decision-centered view of user preference

Let $x \in \mathcal{X}$ denote a decision context, such as a recommendation query, a planning state, a tool-using situation, or a candidate-set question. Let $\mathcal{Y}(x)$ denote the feasible decisions available in that context. A user is modeled not by free-form textual statements alone, but by a conditional decision process

$$P_u(y \mid x), \qquad y \in \mathcal{Y}(x), \tag{1}$$

where $u$ indexes a user or user profile. We interpret $P_u(\cdot \mid x)$ as the user's *decision preference distribution*. In deterministic settings, $P_u$ collapses to a decision rule $f_u(x)$.

The central evaluation target is therefore behavioral:

> After interaction, does the assistant choose as the user would choose on new contexts?

### 3.2 Interactive learning setting

An assistant agent interacts with a user over $t$ rounds and observes a trajectory

$$\mathcal{D}_t = \{(x_1, y_1), \ldots, (x_t, y_t)\}, \tag{2}$$

where each $y_i$ is the user's chosen decision, ranking, correction, or preference-bearing response for context $x_i$.

The assistant uses an interaction-conditioned learning mechanism $\mathcal{A}_\phi$ to produce an internal adaptation state

$$z_t = \mathcal{A}_\phi(\mathcal{D}_t), \tag{3}$$

and predicts future user-consistent decisions through

$$Q_\phi(y \mid x, \mathcal{D}_t). \tag{4}$$

Here $Q_\phi$ is the assistant's post-interaction decision policy.

### 3.3 Preference types as train/test units

We assume that each user $u$ belongs to a preference type space $\Theta$, induced by profile descriptions, user metadata, or latent decision patterns. In practice, a dataset may already contain multiple users or user profiles. We interpret each user/profile as one instance of a decision preference type.

A key design principle of this paper is that **train/test splitting is performed over users or profiles**, not only over contexts. The reason is conceptual: the goal of interactive preference learning is to adapt to new preference types. If evaluation only holds out contexts while reusing the same users, it primarily measures within-user interpolation, not cross-preference generalization.

Thus, given a set of users or profiles $\{u_1, \ldots, u_n\}$, we partition them into

$$\mathcal{U}_{\text{train}} \cap \mathcal{U}_{\text{test}} = \varnothing, \qquad \mathcal{U}_{\text{train}} \cup \mathcal{U}_{\text{test}} = \mathcal{U}. \tag{5}$$

The assistant is trained on interactive episodes involving users in $\mathcal{U}_{\text{train}}$, and evaluated on unseen users in $\mathcal{U}_{\text{test}}$.

## 4 From User Simulation to Decision Preference Fitting

### 4.1 What simulation is used for

We do not treat an LLM simulator as a ground-truth model of human psychology. Instead, we use it to instantiate a family of decision-making behaviors under controlled user profiles. The scientific object of interest is not simulator realism per se, but whether an interactive learning mechanism can fit and generalize across a family of decision preference distributions.

Formally, a simulator is a mapping from a profile $p$ and a context $x$ to a user-side decision process

$$P_p(y \mid x), \tag{6}$$

or, in ranking settings, to a distribution over permutations or preference orders over candidate decisions.

The simulator is suitable for evaluation only if two properties hold:

1. **Intra-profile stability:** repeated decisions under the same profile are sufficiently consistent to define an evaluable preference pattern;
2. **Inter-profile diversity:** different profiles induce meaningfully different decision behaviors.

These are benchmark design requirements, not claims about human fidelity.

### 4.2 Behavioral objective

Given interaction data $\mathcal{D}_t$ from a user/profile $u$, the assistant's objective is to fit the user's future decision behavior. A natural population risk is

$$R(\phi; u) = \mathbb{E}_{x \sim \mu_u}\left[\ell\big(Q_\phi(\cdot \mid x, \mathcal{D}_t), P_u(\cdot \mid x)\big)\right], \tag{7}$$

where $\mu_u$ is the context distribution for user $u$ and $\ell$ is a decision-policy discrepancy, such as cross-entropy, total variation, or ranking loss.

In many realistic evaluation settings, the full user distribution $P_u(\cdot \mid x)$ is not directly available, but sampled decisions or rankings are. The empirical benchmark therefore focuses on observable post-interaction alignment between assistant decisions and user decisions.

### 4.3 Post-interaction decision consistency

We define the central benchmark concept as follows.

**Definition 1** (Post-interaction decision consistency)**.** *For a user $u$ and interaction length $t$, the assistant's post-interaction decision consistency is*

$$\mathrm{BDC}_t(u) = \mathbb{E}_{x \sim \mu_u^{\mathrm{test}}} \left[ \mathrm{sim}\big(\hat{y}_\phi(x, \mathcal{D}_t), \hat{y}_u(x)\big) \right], \tag{8}$$

*where $\hat{y}_\phi(x, \mathcal{D}_t)$ is the assistant's decision on a held-out context after interacting with the user for $t$ rounds, $\hat{y}_u(x)$ is the user's decision on the same held-out context, and* sim *is a decision-similarity function such as exact match, top-$K$ overlap, or rank correlation.*

This definition captures the benchmark target directly. The assistant succeeds if, after interaction, it makes the decisions the user would make.

## 5 Preference-Type Construction and Validation

### 5.1 Profiles as decision preference types

A profile can be specified by explicit attributes (e.g., budget sensitivity, risk attitude, preference for concise answers, tool conservatism, or planning style), by user identifiers already present in a dataset, or by richer LLM-generated role descriptions. The critical requirement is not that the textual profile be semantically elegant, but that it induce a distinct and measurable decision pattern.

Concretely, we instantiate a set of profiles

$$\mathcal{P} = \{p_1, \ldots, p_M\}, \tag{9}$$

and use an LLM user agent conditioned on $p_m$ to generate decisions across contexts. Each $p_m$ is treated as a preference type.

### 5.2 Why diversity must be verified

If different profiles do not actually yield different decisions, then splitting by profile is meaningless and any apparent generalization result is weak. Conversely, if profiles induce unstable or noisy decisions, then the benchmark target itself is ill-posed. We therefore require explicit validation of both diversity and stability before using the simulated users for interactive preference-learning evaluation.

### 5.3 Validation criteria

For each profile $p$, let $f_p(x)$ denote the profile-conditioned decision rule or mean decision under repeated sampling. We recommend the following validation checks.

**(1) Intra-profile stability.** Repeatedly query the same profile on the same context or matched contexts and compute the self-consistency rate

$$\mathrm{Stab}(p) = \mathbb{E}_x \left[ \mathbb{1}\big(f_p^{(1)}(x) = f_p^{(2)}(x)\big) \right], \tag{10}$$

or an analogous ranking-consistency statistic.

**(2) Inter-profile disagreement.** Measure average pairwise decision disagreement

$$\mathrm{Div}(p_i, p_j) = \mathbb{E}_x \left[ 1 - \mathrm{sim}\big(f_{p_i}(x), f_{p_j}(x)\big) \right]. \tag{11}$$

The benchmark should include profiles with substantial pairwise diversity.

**(3) Clusterability in behavior space.** Represent each profile by its decision vector or ranking statistics across a probe set of contexts, and verify that profiles are separable in behavior space by clustering, nearest-neighbor retrieval, or contrastive classification.

**(3b) Coverage radius in behavior space.** Beyond pairwise diversity or clusterability, we recommend measuring how well the training profile set covers the behavior space induced by a probe set of contexts. Let $b_p$ denote the behavior signature of profile $p$, and define the coverage radius of a training profile set $S$ by

$$\rho(S) = \sup_{p \in \mathcal{P}} \min_{q \in S} d\big(b_p, b_q\big), \tag{12}$$

where $d$ is a distance on behavior signatures, such as average total variation over probe contexts. Under a Lipschitz transfer assumption, smaller coverage radius yields a tighter worst-case bound on cross-profile generalization error; see Appendix **??**. This makes coverage a more directly relevant notion of profile diversity than pairwise dispersion alone.

**(4) Coverage of decision trade-offs.** Ensure that the profile set spans multiple recurring trade-offs relevant to the application domain, such as speed vs. quality, risk vs. reward, minimal vs. proactive assistance, or conservative vs. exploratory tool use.

Only after these checks should the profile family be used as an evaluation bed for interactive preference learning.

# 6 Training and Evaluation Protocol

## 6.1 Episode structure

Each interactive episode proceeds in two phases.

**Adaptation phase.** The assistant interacts with a user agent instantiated from profile $u$ for $t$ rounds. At each round, the assistant may ask questions, propose actions, receive corrections, or observe the user's chosen decision.

**Evaluation phase.** After interaction, the assistant is tested on a held-out set of contexts for the same user/profile. On each held-out context, the user agent produces a decision or ranking over candidate actions, and the assistant produces its own decision or ranking. The two are compared using a fixed decision-similarity metric.

## 6.2 Train/validation/test split

We recommend the following split strategy.

1. Partition users/profiles into train, validation, and test groups.
2. Use disjoint context sets within each user/profile for adaptation and held-out evaluation.
3. Report final results only on unseen users/profiles in the test split.

This protocol distinguishes three capabilities:

- **within-user fitting:** can the agent fit a user after observing interaction?
- **across-user generalization:** can the agent adapt to unseen preference types?
- **sample efficiency:** how quickly does post-interaction decision consistency improve with interaction length?

## 6.3 Interactive learning objective

The framework is compatible with many learning mechanisms: recurrent adaptation, memory-based adaptation, latent-state inference, in-context learning, and gradient-based meta-learning. A generic training objective is

$$\mathcal{L}_{\text{train}}(\phi) = \mathbb{E}_{u \sim \mathcal{U}_{\text{train}}} \mathbb{E}_{\mathcal{D}_t \sim u} \mathbb{E}_{x \sim \mu_u^{\text{query}}} \Big[ -\log Q_\phi(y_u \mid x, \mathcal{D}_t) \Big], \tag{13}$$

where $y_u$ is the user decision on a query context. In ranking settings, this can be replaced or augmented by listwise or pairwise ranking losses.

**Algorithm 1** Evaluation of Interactive Preference Learning via Decision Consistency

**Require:** Profile set $\mathcal{U}$, assistant agent $Q_\phi$, interaction budget $t$

Split $\mathcal{U}$ into $\mathcal{U}_{\text{train}}, \mathcal{U}_{\text{val}}, \mathcal{U}_{\text{test}}$
Validate profile stability and inter-profile diversity on a probe set
**for** each training iteration **do**
    Sample user/profile $u \sim \mathcal{U}_{\text{train}}$
    Interact for $t$ rounds to obtain $\mathcal{D}_t$
    Sample held-out query contexts for $u$
    Update $\phi$ using user decisions or rankings on query contexts
**end for**
**for** each test user/profile $u \in \mathcal{U}_{\text{test}}$ **do**
    Interact for $t$ rounds to obtain $\mathcal{D}_t$
    Evaluate assistant decisions on held-out contexts
    Compute exact-match, top-$K$, and ranking similarity against user decisions
**end for**
Report aggregated post-interaction decision consistency and adaptation curves

# 7 Evaluation Metrics

We recommend a metric suite centered on observable decision behavior.

## 7.1 Exact decision agreement

When both assistant and user output a single decision,

$$\text{Acc}_{\text{dec}}(u) = \mathbb{E}_{x \sim \mu_u^{\text{test}}} \left[ \mathbb{1}\big(\hat{y}_\phi(x, \mathcal{D}_t) = \hat{y}_u(x)\big) \right]. \tag{14}$$

This is the most direct measure of whether the assistant decides as the user would decide.

## 7.2 Top-$K$ overlap

When decisions are rankings or ranked candidate sets, define

$$\text{TopK}(u) = \mathbb{E}_{x \sim \mu_u^{\text{test}}} \left[ \frac{|\text{TopK}(\pi_\phi(x, \mathcal{D}_t)) \cap \text{TopK}(\pi_u(x))|}{K} \right]. \tag{15}$$

This metric is useful when multiple top choices are behaviorally plausible.

## 7.3 Rank correlation

For full rankings, compute Kendall's $\tau$ or Spearman's $\rho$ between the assistant ranking and the user ranking. This captures whether the assistant reproduces the user's ordering over candidate decisions.

## 7.4 Adaptation curve

To measure sample efficiency, report

$$\text{AdaptCurve}(t) = \mathbb{E}_{u \in \mathcal{U}_{\text{test}}}[\text{BDC}_t(u)], \tag{16}$$

as a function of the number of interaction rounds. This reveals how quickly a learning mechanism acquires user-consistent decision behavior.

## 7.5 Cross-profile generalization gap

To isolate generalization beyond memorization, compare performance on seen and unseen users/profiles:

$$\text{GenGap} = \mathbb{E}_{u \in \mathcal{U}_{\text{train}}}[\text{BDC}_t(u)] - \mathbb{E}_{u \in \mathcal{U}_{\text{test}}}[\text{BDC}_t(u)]. \tag{17}$$

Smaller generalization gaps indicate stronger adaptation to unseen preference types.

## 8 Theoretical Properties

We now clarify what post-interaction decision consistency does and does not identify. The main message is that exact-match decision consistency is a direct objective for deterministic preference fitting, but stochastic preference fitting requires a proper-scoring extension. We also quantify how simulator instability affects estimation, and how profile-space coverage controls cross-profile generalization.

**Proposition 1** (Directness of exact-match decision consistency)**.** *Assume that for each user/profile $u$ there exists a latent canonical decision rule*

$$y_u^\star(x) =_{y \in \mathcal{Y}(x)} r_u(x, y), \tag{18}$$

*which is unique almost surely under the test context distribution. Let*

$$\mathrm{BDC}_t^\star(u) = \mathbb{E}_{x \sim \mu_u^{\mathrm{test}}} \left[ \mathbb{1}\big(\hat{y}_\phi(x, \mathcal{D}_t) = y_u^\star(x)\big) \right]. \tag{19}$$

*Then the post-interaction decision error*

$$L_{\mathrm{dec}}(t, u) = \mathbb{E}_{x \sim \mu_u^{\mathrm{test}}} \left[ \mathbb{1}\big(\hat{y}_\phi(x, \mathcal{D}_t) \neq y_u^\star(x)\big) \right] \tag{20}$$

*satisfies*

$$L_{\mathrm{dec}}(t, u) = 1 - \mathrm{BDC}_t^\star(u). \tag{21}$$

*Hence maximizing exact-match decision consistency is equivalent to minimizing decision disagreement with the user's canonical decision rule.*

**Proposition 2** (Why proxy metrics require stronger assumptions)**.** *Let $h_u(x, y)$ denote the conditional expectation of a proxy signal, such as satisfaction, self-report alignment, or free-form judged quality, when decision $y$ is taken in context $x$ for user $u$. A proxy metric is Fisher-consistent for decision fitting only if, for almost every $x$, the ordering induced by $h_u(x, \cdot)$ agrees with the ordering induced by the latent decision utility $r_u(x, \cdot)$. Without such order calibration, optimizing the proxy can select a decision that differs from $y_u^\star(x)$.*

**Proposition 3** (What exact-match decision consistency identifies under stochastic preferences)**.** *If the user's behavior is stochastic, with $y \sim P_u(\cdot \mid x)$, then exact-match decision consistency can be written as*

$$\mathrm{BDC}_t^{\mathrm{EM}}(u) = \mathbb{E}_{x \sim \mu_u^{\mathrm{test}}} \left[ P_u\big(\hat{y}_\phi(x, \mathcal{D}_t) \mid x\big) \right]. \tag{22}$$

*Therefore, exact-match decision consistency identifies the Bayes action (the modal decision) but not the full conditional decision distribution unless $P_u(\cdot \mid x)$ is degenerate almost surely.*

**Theorem 1** (Distributional identification via a proper-scoring extension)**.** *Suppose repeated user or simulator queries allow us to evaluate the assistant's predictive distribution $Q_\phi(\cdot \mid x, \mathcal{D}_t)$ with the log score*

$$\mathrm{BDC}_t^{\log}(u) = \mathbb{E}_{x \sim \mu_u^{\mathrm{test}}} \mathbb{E}_{y \sim P_u(\cdot \mid x)} \left[ \log Q_\phi(y \mid x, \mathcal{D}_t) \right]. \tag{23}$$

*Then*

$$\mathrm{BDC}_t^{\log}(u) = -\mathbb{E}_{x \sim \mu_u^{\mathrm{test}}} \left[ H\big(P_u(\cdot \mid x)\big) + \mathrm{KL}\big(P_u(\cdot \mid x) \,\|\, Q_\phi(\cdot \mid x, \mathcal{D}_t)\big) \right]. \tag{24}$$

*Consequently, $\mathrm{BDC}_t^{\log}(u)$ is maximized if and only if*

$$Q_\phi(\cdot \mid x, \mathcal{D}_t) = P_u(\cdot \mid x) \qquad \text{for } \mu_u^{\mathrm{test}}\text{-almost every } x. \tag{25}$$

**Theorem 2** (Simulator instability yields additive estimation bias)**.** *Let $\tilde{y}_u(x)$ denote the observed simulator decision and $y_u^\star(x)$ the latent canonical decision. Define the pointwise simulator error rate*

$$\eta_u(x) = \mathbb{P}\big(\tilde{y}_u(x) \neq y_u^\star(x) \mid x, u\big), \tag{26}$$

*and let $\bar{\eta}_u = \mathbb{E}_{x \sim \mu_u^{\mathrm{test}}}[\eta_u(x)]$. Then*

$$\left| \mathrm{BDC}_t^{\mathrm{sim}}(u) - \mathrm{BDC}_t^\star(u) \right| \leq \bar{\eta}_u. \tag{27}$$

*Moreover, if $\widehat{\mathrm{BDC}}_{t,n}^{\mathrm{sim}}(u)$ is the empirical estimate computed from $n$ i.i.d. held-out contexts, then with probability at least $1 - \delta$,*

$$\left| \widehat{\mathrm{BDC}}_{t,n}^{\mathrm{sim}}(u) - \mathrm{BDC}_t^\star(u) \right| \leq \bar{\eta}_u + \sqrt{\frac{\log(2/\delta)}{2n}}. \tag{28}$$

| Benchmark | Profile signal | Native action space | Decision unit used in this paper | Role |
|---|---|---|---|---|
| PersonaLens | Rich user profiles with preferences and interaction histories | Task-oriented assistant responses over diverse user-task scenarios | Candidate-set decision over response/action plans derived from the same task context | Main benchmark |
| ETAPP | Explicit user profiles and tool-utilizing preferences | Personalized tool invocation in a sandbox environment | Tool/API selection together with argument configuration | Main benchmark |
| ToolSpectrum | Explicit user profiles and environmental factors | Personalized tool selection under overlapping tool sets | Tool/API selection together with personalized and environment-dependent arguments | Main benchmark |

Table 1: Primary benchmarks used in the main experimental section. All three benchmarks provide profile-conditioned agent settings that are directly compatible with post-interaction decision consistency evaluation.

**Theorem 3** (Profile-space coverage controls worst-case cross-profile generalization)**.** *Let*

$$G_t(u) = 1 - \mathrm{BDC}_t^{\star}(u) \tag{29}$$

*denote the post-interaction decision error for profile $u$. Let $b(u)$ denote the profile's behavior signature on a probe set of contexts, and let $d$ be a metric on behavior signatures. Assume that for some $L > 0$,*

$$|G_t(u) - G_t(v)| \le L\, d\big(b(u), b(v)\big) \qquad \textit{for all } u, v. \tag{30}$$

*For a training profile set $S$, define the coverage radius*

$$\rho(S) = \sup_{u \in \mathrm{supp}(\mathcal{P})} \min_{v \in S} d\big(b(u), b(v)\big). \tag{31}$$

*If*

$$G_t(v) \le \varepsilon_{\mathrm{train}} \qquad \textit{for all } v \in S, \tag{32}$$

*then*

$$\sup_{u \in \mathrm{supp}(\mathcal{P})} G_t(u) \le \varepsilon_{\mathrm{train}} + L\rho(S). \tag{33}$$

*Thus, under smooth transfer in behavior space, the relevant notion of profile diversity is coverage: broader behavioral coverage leads to a tighter worst-case generalization bound.*

Full proofs are given in Appendix **??**.

# 9 Experimental Design

## 9.1 Benchmark selection

To keep the empirical study tightly aligned with personalized LLM agents, we restrict the main experiments to benchmarks that expose explicit or semi-explicit user profiles and require agentic decisions in task-oriented settings. We therefore use PersonaLens (**?**), ETAPP (**?**), and ToolSpectrum (**?**) as the primary benchmarks, and exclude generic personalization datasets without an agent environment from the main experimental section.

## 9.2 Decision-space instantiation

A central requirement of our framework is that the assistant and the user be compared in a common decision space after interaction. For ETAPP and ToolSpectrum, this decision space is native: the assistant must choose a tool/API and, when necessary, its arguments. We therefore treat the final tool invocation tuple as the atomic decision.

PersonaLens is originally evaluated through open-ended personalized task completion rather than discrete decision matching. To make it compatible with our framework, we convert each held-out task into a candidate-set decision problem. For each task context $x$, we use a fixed held-out proposer model to generate $K = 5$ semantically distinct candidate response or action plans. These candidates are

| Method family | Description |
|---|---|
| Non-adaptive assistant | Receives only the current task context and never conditions on user-specific evidence. |
| Profile-text baseline | Receives the explicit profile text but no interaction history from the current episode. |
| Raw-history ICL | Conditions on the full support trajectory by direct prompt concatenation. |
| Retrieval-augmented ICL | Retrieves the most relevant support cases for the current query before in-context inference. |
| Latent-state adaptation | Encodes the support trajectory into an explicit user state or memory vector, which conditions subsequent decisions. |
| PEFT online adaptation | Performs lightweight parameter updates (e.g., LoRA or prefix tuning) on support interactions before query-time decision making. |
| Meta-learned adaptation | Trains the interaction mechanism episodically so that it can adapt rapidly to unseen profiles from a small number of support interactions. |
| Oracle-profile upper bound | Receives gold profile attributes and full user preference specifications when available; used only as an upper bound. |

Table 2: Baseline families used across all primary benchmarks. The comparison isolates whether explicit interaction-conditioned adaptation improves post-interaction decision consistency beyond profile text alone or raw history prompting.

anonymized and randomly permuted. The user agent then selects or ranks the candidates, producing the target user decision, while the evaluated assistant selects or ranks candidates after interacting with the same user profile. This procedure preserves the original task semantics while yielding exact-match, top-$K$, and rank-based decision comparisons.

### 9.3 Preference types, episodes, and splits

We treat each user profile, or each profile template when persistent user identities are not provided, as one preference type. No profile is allowed to appear in more than one split.

For PersonaLens and ToolSpectrum, we use profile-level train/validation/test splits with a 70/10/20 ratio. For ETAPP, where the number of distinct users is smaller, we use leave-two-users-out cross-validation and report the mean across folds. Within each profile, task instances are partitioned into disjoint support and query sets.

Each evaluation episode has two phases. In the adaptation phase, the assistant interacts with a user for $t$ support instances and observes behavior-bearing signals such as choices, corrections, rankings, or tool preferences. In the evaluation phase, the assistant is tested on held-out query instances from the same profile, and its decisions are compared against the user's decisions on the same instances. We report results for interaction budgets $t \in \{0, 1, 2, 4, 8\}$ to characterize sample-efficient adaptation.

For ToolSpectrum, we report two generalization settings: (i) *profile generalization*, where train and test differ in user profiles but share the same environment distribution, and (ii) *profile×environment generalization*, where train and test differ in both user profiles and environmental conditions. This separates adaptation to user preference from robustness to changing execution context.

### 9.4 Assistant backbones and adaptation methods

We instantiate all methods with three assistant backbones: Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct, and GPT-4o. We evaluate each backbone under the same profile-level episode construction so that differences can be attributed to adaptation mechanisms rather than to split design.

### 9.5 Primary and auxiliary metrics

Our primary metrics are the decision-level metrics defined earlier in the paper. Specifically, we report:

1. **Exact decision agreement** between assistant and user decisions on held-out contexts;
2. **Top-$K$ overlap** when multiple candidate decisions are behaviorally plausible;
3. **Rank correlation** when the user and assistant output full or partial rankings;
4. **Adaptation curves** as a function of the interaction budget $t$;

5. **Cross-profile generalization gap** between seen and unseen preference types.

For ETAPP and ToolSpectrum, we additionally report tool-level exact match and argument-level matching scores, since these are natural behavioral readouts of personalized tool use. For PersonaLens, we report the benchmark's original judge-based personalization and task-success metrics only as auxiliary indicators. They are not treated as the primary target of evaluation, because our main claim concerns post-interaction decision consistency rather than free-form judged quality.

### 9.6 Main hypotheses

We evaluate the following hypotheses:

**H1:** Explicit interaction-conditioned adaptation improves post-interaction decision consistency relative to non-adaptive and profile-text baselines.

**H2:** Profile-level train/test splitting is substantially more demanding than context-only splitting and yields a more faithful estimate of personalization generalization.

**H3:** Methods with explicit adaptation mechanisms outperform raw-history prompting on unseen profiles, especially under small interaction budgets.

**H4:** Personalized tool-use benchmarks and personalized assistant benchmarks can be evaluated in a unified way once both are cast into a common decision space.

**H5:** On ToolSpectrum, joint variation in user profile and environment reveals failure modes that are not visible under profile-only evaluation.

### 9.7 Ablations

We conduct the following ablations to isolate what drives performance:

1. **Interaction budget ablation:** vary $t \in \{0, 1, 2, 4, 8\}$ and report adaptation curves.
2. **Split ablation:** compare profile-level train/test splits against context-only splits to quantify how much context-only evaluation overestimates personalization capability.
3. **Profile-information ablation:** compare using profile text only, support behavior only, and both together.
4. **History representation ablation:** compare raw concatenation, retrieval-based history selection, summary-based memory, and latent-state adaptation.
5. **PersonaLens candidate-set ablation:** vary the number of candidate actions $K$ and regenerate candidate sets with different held-out proposer models to verify that conclusions are not artifacts of a particular proposal pool.
6. **ToolSpectrum environment ablation:** evaluate with and without environmental factors to separate user-specific adaptation from environment-sensitive tool reasoning.
7. **Argument-space complexity ablation:** increase the number of feasible tools or parameter combinations in ETAPP and ToolSpectrum to test whether methods still recover user-specific decision patterns under harder action spaces.

### 9.8 Implementation details

Unless otherwise specified, all hyperparameters are selected on the validation split and then fixed for test evaluation. For all benchmarks, we average results over multiple random seeds for support/query sampling. In PersonaLens, candidate generation is performed by a held-out proposer model that is never used as an evaluated assistant backbone. In ETAPP and ToolSpectrum, all tool decisions are executed in the benchmark-defined environment, and all reported decision metrics are computed on held-out instances only.

## 10 Analysis and Reporting Standards

### 10.1 What counts as success

A method should be considered successful only if it satisfies all of the following:

1. It achieves high post-interaction decision consistency on held-out contexts.
2. It generalizes to unseen users/profiles rather than only to seen ones.
3. It improves consistently with interaction budget.
4. Its gains persist under profile-diversity validation and are not artifacts of profile collapse or simulator instability.

### 10.2 Statistical reporting

For all metrics, we recommend:

- mean $\pm$ standard error across users/profiles,
- bootstrap confidence intervals,
- paired significance tests for adaptation gains,
- per-profile breakdowns to expose heterogeneity,
- correlation analysis between profile-diversity measures and adaptation performance.

### 10.3 Failure analysis

We further recommend categorizing failure cases into:

1. **preference confusion:** the assistant predicts the wrong trade-off direction;
2. **under-adaptation:** the assistant remains close to a generic policy;
3. **overfitting to early interaction:** the assistant overcommits to noisy early decisions;
4. **profile collapse:** distinct profiles lead to similar user decisions, weakening the benchmark itself.

## 11 Discussion

**Why this framing is useful.** The proposed framework establishes a unified target across heterogeneous personalized-agent tasks. Instead of comparing incomparable proxies such as satisfaction, verbal preference matching, or simulator agreement in free-form language, we compare what ultimately matters for personalized decision-making: whether the assistant chooses as the user would choose after interaction. These empirical and theoretical properties together justify decision consistency as a direct target for evaluating interactive preference fitting, rather than merely as another benchmark-specific score.

**What this paper does not claim.** We do not claim that LLM simulators are perfect substitutes for humans. We also do not claim that a textual profile exhaustively captures a user's true internal preference. The claim is narrower and operational: if a simulator induces stable and diverse decision patterns, then it can be used to evaluate whether an interaction mechanism can fit and generalize across a class of decision preference distributions.

**Why behavior should dominate self-report.** In real systems, users often provide incomplete, inconsistent, or socially filtered verbal descriptions of what they want. Yet they still reveal stable operative preferences through repeated decisions. A behavior-centered evaluation is therefore both more objective and more directly aligned with deployment needs.

## 12 Limitations

Our framework inherits the limitations of the simulated profile family. If profiles do not cover the diversity of decision patterns relevant to the target application, then benchmark conclusions may be narrow. The quality of evaluation also depends on simulator stability: if user-side decisions are too noisy, preference fitting becomes ill-posed. Finally, some domains require long-horizon, sequential, or multi-step preferences that are not fully captured by one-shot decisions or rankings. Extending the framework to richer sequential preference structures is an important direction for future work.

## 13 Conclusion

This paper proposes a behavior-centered evaluation framework for interactive preference learning. The key idea is simple: in personalized decision-making, the most operationally faithful reflection of preference is not what the user says, but how the user decides. We therefore model user preference as a decision distribution, use profiles or users as preference-type units, split training and testing by these units, and evaluate post-interaction decision consistency between the assistant and the user on held-out contexts. By validating profile diversity and measuring exact decision agreement, top-$K$ overlap, ranking similarity, and sample-efficient adaptation, the framework provides a unified and objective basis for comparing interactive preference-learning methods. The resulting question is precise and practically important: can a personalized assistant learn, from interaction, to decide the way different users decide?

## A Theoretical Properties and Proofs

### A.1 Auxiliary setup

We distinguish between a latent canonical decision rule and the possibly unstable decisions produced by a simulator. For a user/profile $u$ and context $x$, let $r_u(x, y)$ denote the latent decision utility of action $y \in \mathcal{Y}(x)$. When the maximizer is unique, define the canonical decision

$$y_u^\star(x) =_{y \in \mathcal{Y}(x)} r_u(x, y). \tag{34}$$

Let $\tilde{y}_u(x)$ denote the observed simulator-side decision, which may be a noisy realization of the canonical decision. Let $\hat{y}_\phi(x, \mathcal{D}_t)$ denote the assistant's post-interaction decision after observing the support trajectory $\mathcal{D}_t$.

For stochastic preference modeling, we write $P_u(\cdot \mid x)$ for the user's conditional decision distribution and $Q_\phi(\cdot \mid x, \mathcal{D}_t)$ for the assistant's predictive distribution.

To relate diversity to generalization, let $\mathcal{X}_{\text{probe}} = \{x_1, \ldots, x_m\}$ be a fixed probe set of contexts and define the behavior signature

$$b(u) = \Big(P_u(\cdot \mid x_1), \ldots, P_u(\cdot \mid x_m)\Big). \tag{35}$$

A natural distance between two behavior signatures is the average total variation distance

$$d\big(b(u), b(v)\big) = \frac{1}{m} \sum_{j=1}^{m} \mathrm{TV}\Big(P_u(\cdot \mid x_j), P_v(\cdot \mid x_j)\Big). \tag{36}$$

**Assumption 1** (Canonical decision rule)**.** *For each profile $u$, there exists a latent decision utility $r_u(x, y)$ such that $y_u^\star(x) =_{y \in \mathcal{Y}(x)} r_u(x, y)$ exists and is unique for $\mu_u^{\text{test}}$-almost every $x$.*

**Assumption 2** (Evaluation sampling)**.** *Held-out contexts $x_1, \ldots, x_n$ are sampled i.i.d. from $\mu_u^{\text{test}}$. When repeated simulator queries are used on the same $(u, x)$, the resulting simulator outputs are conditionally independent given $(u, x)$.*

**Assumption 3** (Smooth transfer in behavior space)**.** *For each interaction budget $t$, the post-interaction decision error*

$$G_t(u) = 1 - \mathrm{BDC}_t^\star(u) \tag{37}$$

*is $L$-Lipschitz with respect to the behavior-signature metric $d$, i.e.,*

$$|G_t(u) - G_t(v)| \le L\, d\big(b(u), b(v)\big) \qquad \textit{for all } u, v. \tag{38}$$

### A.2 Proofs

**Proof of Proposition ??.**

*Proof.* Under Assumption **??**, the canonical decision $y_u^\star(x)$ is uniquely defined for almost every $x$. By definition,

$$\mathrm{BDC}_t^\star(u) = \mathbb{E}_{x \sim \mu_u^{\text{test}}} \left[\mathbb{1}\big(\hat{y}_\phi(x, \mathcal{D}_t) = y_u^\star(x)\big)\right]. \tag{39}$$

The corresponding decision error is

$$L_{\mathrm{dec}}(t,u) = \mathbb{E}_{x\sim\mu_u^{\mathrm{test}}}\left[\mathbb{1}\big(\hat{y}_\phi(x,\mathcal{D}_t) \neq y_u^\star(x)\big)\right]. \tag{40}$$

Since

$$\mathbb{1}\big(\hat{y}_\phi(x,\mathcal{D}_t) \neq y_u^\star(x)\big) = 1 - \mathbb{1}\big(\hat{y}_\phi(x,\mathcal{D}_t) = y_u^\star(x)\big), \tag{41}$$

taking expectations yields

$$L_{\mathrm{dec}}(t,u) = 1 - \mathrm{BDC}_t^\star(u). \tag{42}$$

Hence maximizing $\mathrm{BDC}_t^\star(u)$ is exactly equivalent to minimizing disagreement with the user's canonical decision rule. □

**Proof of Proposition ??.**

*Proof.* Let

$$h_u(x,y) = \mathbb{E}[Z_u(x,y) \mid x,u] \tag{43}$$

denote the conditional expectation of some proxy signal $Z_u$, such as satisfaction, self-report agreement, or judged response quality. Suppose there exist a profile $u$, a context $x$, and two feasible actions $y_1, y_2 \in \mathcal{Y}(x)$ such that

$$r_u(x,y_1) > r_u(x,y_2) \qquad \text{but} \qquad h_u(x,y_1) < h_u(x,y_2). \tag{44}$$

Then the decision that maximizes the proxy score at $(u,x)$ selects $y_2$, while the decision that maximizes the latent decision utility selects $y_1$. Therefore, the proxy-optimal policy disagrees with the canonical decision rule on that context. This proves that, absent order preservation between $h_u(x,\cdot)$ and $r_u(x,\cdot)$, optimizing the proxy need not recover the decision-optimal policy.

Equivalently, a proxy metric can be Fisher-consistent for preference fitting only if its conditional expectation preserves the action ordering induced by the latent decision utility almost everywhere. □

**Proof of Proposition ??.**

*Proof.* Suppose that for each context $x$, the user draws a decision $y$ from the distribution $P_u(\cdot \mid x)$, and the assistant outputs a deterministic decision $\hat{y}_\phi(x,\mathcal{D}_t)$. Then

$$\mathrm{BDC}_t^{\mathrm{EM}}(u) = \mathbb{E}_{x\sim\mu_u^{\mathrm{test}}}\mathbb{E}_{y\sim P_u(\cdot|x)}\left[\mathbb{1}\big(\hat{y}_\phi(x,\mathcal{D}_t) = y\big)\right] \tag{45}$$

$$= \mathbb{E}_{x\sim\mu_u^{\mathrm{test}}}\left[P_u\big(\hat{y}_\phi(x,\mathcal{D}_t) \mid x\big)\right]. \tag{46}$$

For a fixed $x$, the quantity $P_u(\hat{y} \mid x)$ is maximized by any action

$$\hat{y} \in_{y\in\mathcal{Y}(x)} P_u(y \mid x). \tag{47}$$

Thus exact-match decision consistency identifies the Bayes action, i.e. the modal decision under $P_u(\cdot \mid x)$. However, if $P_u(\cdot \mid x)$ is not degenerate, then many distinct conditional distributions can induce the same mode. Therefore exact-match decision consistency does not identify the full stochastic decision distribution unless $P_u(\cdot \mid x)$ collapses to a point mass almost surely. □

**Proof of Theorem ??.**

*Proof.* By definition,

$$\mathrm{BDC}_t^{\log}(u) = \mathbb{E}_{x\sim\mu_u^{\mathrm{test}}}\mathbb{E}_{y\sim P_u(\cdot|x)}\left[\log Q_\phi(y \mid x,\mathcal{D}_t)\right] \tag{48}$$

$$= \mathbb{E}_{x\sim\mu_u^{\mathrm{test}}}\left[\sum_{y\in\mathcal{Y}(x)} P_u(y \mid x)\log Q_\phi(y \mid x,\mathcal{D}_t)\right]. \tag{49}$$

For each fixed $x$, the standard entropy–KL identity gives

$$\sum_y P_u(y \mid x)\log Q_\phi(y \mid x,\mathcal{D}_t) = -H\big(P_u(\cdot \mid x)\big) - \mathrm{KL}\big(P_u(\cdot \mid x)\,\|\,Q_\phi(\cdot \mid x,\mathcal{D}_t)\big). \tag{50}$$

Taking expectation over $x$ yields

$$\mathrm{BDC}_t^{\log}(u) = -\mathbb{E}_{x\sim\mu_u^{\mathrm{test}}}\left[H\big(P_u(\cdot \mid x)\big) + \mathrm{KL}\big(P_u(\cdot \mid x)\,\|\,Q_\phi(\cdot \mid x,\mathcal{D}_t)\big)\right]. \tag{51}$$

The entropy term depends only on the user distribution and not on the assistant. Therefore $\mathrm{BDC}_t^{\log}(u)$ is maximized if and only if the KL divergence vanishes for $\mu_u^{\mathrm{test}}$-almost every $x$, which holds exactly when

$$Q_\phi(\cdot \mid x, \mathcal{D}_t) = P_u(\cdot \mid x) \qquad \mu_u^{\mathrm{test}}\text{-a.s.} \tag{52}$$

□

**Proof of Theorem ??.**

*Proof.* Define

$$\mathrm{BDC}_t^{\mathrm{sim}}(u) = \mathbb{E}_{x\sim\mu_u^{\mathrm{test}}}\mathbb{E}\left[\mathbb{1}\big(\hat{y}_\phi(x,\mathcal{D}_t) = \tilde{y}_u(x)\big) \,\middle|\, x, u\right]. \tag{53}$$

For each $(u, x)$, we have the pointwise inequality

$$\left|\mathbb{1}\big(\hat{y}_\phi(x,\mathcal{D}_t) = \tilde{y}_u(x)\big) - \mathbb{1}\big(\hat{y}_\phi(x,\mathcal{D}_t) = y_u^\star(x)\big)\right| \le \mathbb{1}\big(\tilde{y}_u(x) \neq y_u^\star(x)\big). \tag{54}$$

Taking conditional expectation given $(x, u)$ gives

$$\left|\mathbb{E}\big[\mathbb{1}\big(\hat{y}_\phi(x,\mathcal{D}_t) = \tilde{y}_u(x)\big) - \mathbb{1}\big(\hat{y}_\phi(x,\mathcal{D}_t) = y_u^\star(x)\big) \,\middle|\, x, u\big]\right| \le \eta_u(x). \tag{55}$$

Taking expectation over $x \sim \mu_u^{\mathrm{test}}$ yields

$$\left|\mathrm{BDC}_t^{\mathrm{sim}}(u) - \mathrm{BDC}_t^\star(u)\right| \le \bar{\eta}_u. \tag{56}$$

Now let

$$\widehat{\mathrm{BDC}}_{t,n}^{\mathrm{sim}}(u) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}\big(\hat{y}_\phi(x_i,\mathcal{D}_t) = \tilde{y}_u(x_i)\big), \tag{57}$$

where $x_1, \ldots, x_n$ are i.i.d. held-out contexts. By Assumption **??**, the summands are i.i.d. in $[0, 1]$, so Hoeffding's inequality implies that with probability at least $1 - \delta$,

$$\left|\widehat{\mathrm{BDC}}_{t,n}^{\mathrm{sim}}(u) - \mathrm{BDC}_t^{\mathrm{sim}}(u)\right| \le \sqrt{\frac{\log(2/\delta)}{2n}}. \tag{58}$$

Finally, by the triangle inequality,

$$\left|\widehat{\mathrm{BDC}}_{t,n}^{\mathrm{sim}}(u) - \mathrm{BDC}_t^\star(u)\right| \le \left|\widehat{\mathrm{BDC}}_{t,n}^{\mathrm{sim}}(u) - \mathrm{BDC}_t^{\mathrm{sim}}(u)\right| + \left|\mathrm{BDC}_t^{\mathrm{sim}}(u) - \mathrm{BDC}_t^\star(u)\right|, \tag{59}$$

which proves

$$\left|\widehat{\mathrm{BDC}}_{t,n}^{\mathrm{sim}}(u) - \mathrm{BDC}_t^\star(u)\right| \le \bar{\eta}_u + \sqrt{\frac{\log(2/\delta)}{2n}} \tag{60}$$

with probability at least $1 - \delta$. □

**Corollary 1** (Repeated-query stabilization)**.** *Suppose that on a fixed $(u, x)$, repeated simulator queries are i.i.d., and each query returns an incorrect decision with probability at most $\eta < 1/2$. Let $\bar{y}_u^{(m)}(x)$ be the majority vote of $m$ repeated simulator outputs. Then*

$$\mathbb{P}\big(\bar{y}_u^{(m)}(x) \neq y_u^\star(x)\big) \le \exp\Big(-2m\big(\tfrac{1}{2} - \eta\big)^2\Big). \tag{61}$$

*Therefore, if majority-vote labels are used in place of single simulator draws, the instability term in Theorem **??** decays exponentially in the number of repeated queries.*

*Proof.* Let $E_i$ be the indicator that the $i$-th simulator query is incorrect. Then $E_1, \ldots, E_m$ are i.i.d. Bernoulli with mean at most $\eta$. A majority vote is wrong only if

$$\frac{1}{m}\sum_{i=1}^{m} E_i \ge \frac{1}{2}. \tag{62}$$

Applying Hoeffding's inequality,

$$\mathbb{P}\left(\frac{1}{m}\sum_{i=1}^{m} E_i - \eta \ge \frac{1}{2} - \eta\right) \le \exp\Big(-2m\big(\tfrac{1}{2} - \eta\big)^2\Big). \tag{63}$$

This proves the result. □

**Proof of Theorem ??.**

*Proof.* Fix any profile $u \in \mathrm{supp}(\mathcal{P})$, and let

$$v_u \in_{v \in S} d\big(b(u), b(v)\big) \tag{64}$$

be a nearest training profile in behavior space. By Assumption **??**,

$$G_t(u) \le G_t(v_u) + L\, d\big(b(u), b(v_u)\big). \tag{65}$$

Since $G_t(v) \le \varepsilon_{\mathrm{train}}$ for all $v \in S$, we have

$$G_t(u) \le \varepsilon_{\mathrm{train}} + L\, d\big(b(u), b(v_u)\big) \le \varepsilon_{\mathrm{train}} + L\rho(S). \tag{66}$$

Because this bound holds for every $u \in \mathrm{supp}(\mathcal{P})$, taking the supremum over $u$ yields

$$\sup_{u \in \mathrm{supp}(\mathcal{P})} G_t(u) \le \varepsilon_{\mathrm{train}} + L\rho(S). \tag{67}$$

This shows that, under smooth transfer in behavior space, the worst-case cross-profile generalization error is controlled by the coverage radius of the training profile set. □

### A.3 Interpretation

The propositions and theorems above clarify the scope of our framework.

First, exact-match post-interaction decision consistency is not merely a convenient notation: under a canonical decision-rule assumption, it is exactly the complement of decision disagreement. This gives it a direct identification property that proxy metrics do not enjoy without additional calibration assumptions.

Second, the theory distinguishes deterministic and stochastic preference fitting. Exact-match decision consistency is sufficient to identify the Bayes action, but not the full decision distribution. When the latter is the scientific target, repeated queries together with a proper-scoring extension recover distributional identification.

Third, simulator instability affects evaluation in a quantitatively simple way: it introduces an additive bias term plus standard finite-sample error. This makes repeated-query stabilization a principled benchmark-design tool rather than a heuristic.

Finally, the generalization result shows that the most relevant notion of profile diversity is not pairwise dispersion alone, but coverage of behavior space. This motivates the profile-diversity validation procedures in the main text.

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